

AI Assisted Coding

ASSIGNMENT 7.4

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Question 1: Mutable Default Argument – Function Bug

Task:

Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

Bug: Mutable default argument

```
def add_item(item, items=[]):  
    items.append(item)  
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.

Prompt:

Fix the mutable default argument issue in the function so that each call gets a fresh list.

Code:

```
1  # Bug: Mutable default argument
2  def add_item(item, items=None):
3      if items is None:
4          items = []
5          items.append(item)
6      return items
7  print(add_item(1))
8  print(add_item(2))
```

Output(O/P)

```
[1]
[2]
```

Explanation:

The default list [] was shared between calls.

Default arguments are created only once.

Using None ensures a new list is created each time.

Question 2: Floating-Point Precision Error

Task:

Analyze given code where floating-point comparison fails. Use AI to correct with tolerance.

Bug: Floating point precision issue

```
def check_sum():  
    return (0.1 + 0.2) == 0.3
```

```
print(check_sum())
```

Expected Output: Corrected function

Prompt:

Fix floating point comparison using tolerance.

Code:

```
1  # Bug: Floating point precision issue  
2  def check_sum():  
3      |      return round(0.1 + 0.2, 10) == round(0.3, 10)  
4  print(check_sum())
```

Output(O/P)

```
True
```

Explanation:

Floating numbers are not exact in binary.

Instead of ==, we compare the difference with a small tolerance.

Question 3: Recursion Error – Missing Base Case

Task:

Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix it.

Bug: No base case

```
def countdown(n):  
    print(n)  
    return countdown(n-1)
```

```
countdown(5)
```

Expected Output : Correct recursion with stopping condition.

Prompt:

Add a stopping condition to prevent infinite recursion.

Code:

```
1  # Bug: No base case  
2  def countdown(n):  
3      if n < 0:  
4          return  
5      print(n)  
6      return countdown(n-1)  
7  countdown(5)
```

Output(O/P)

```
5
4
3
2
1
0
```

Explanation:

Without a base case, recursion never stops.
The base case stops when n becomes 0.

Question 4: Dictionary Key Error

Task:

Analyze given code where a missing dictionary key causes error. Use AI to fix it.

Bug: Accessing non-existing key

```
def get_value():  
    data = {"a": 1, "b": 2}  
    return data["c"]
```

```
print(get_value())
```

Expected Output: Corrected with .get() or error handling.

Prompt:

Fix missing key access safely.

Code:

```
1  # Bug: Accessing non-existing key  
2  def get_value():  
3      data = {"a": 1, "b": 2}  
4      return data.get("c", None)  
5  print(get_value())
```

Output(O/P)

```
None
```

Explanation:

Accessing data["c"] causes KeyError.

.get() prevents crashes and allows default value.

Question 5: Infinite Loop – Wrong Condition

Task:

Analyze given code where the loop never ends. Use AI to detect and fix it.

```
# Bug: Infinite loop
def loop_example():
    i = 0
    while i < 5:
        print(i)
```

Expected Output: Corrected loop increments i.

Prompt:

Fix the infinite loop by incrementing the counter.

Code:

```
1  # Bug: Infinite loop
2  def loop_example():
3      i = 0
4      while i < 5:
5          print(i)
6          i += 1
7
8  loop_example()
```

Output(O/P)

```
0
1
2
3
4
```

Explanation:

‘i’ was never incremented.

The loop condition stayed True forever.

Question 6: Unpacking Error – Wrong Variables

Task:

Analyze given code where tuple unpacking fails. Use AI to fix it.

Bug: Wrong unpacking

a, b = (1, 2, 3)

Expected Output: Correct unpacking or using _ for extra values.

Prompt:

Fix tuple unpacking.

Code:

```
1  # Bug: Wrong unpacking
2  a, b, c = (1, 2, 3)
3  print(a, b)
```

Output(O/P)

```
1 2
```

Explanation:

Tuple had three values but only two variables.

Added c to Fix.

Question 7: Mixed Indentation – Tabs vs Spaces

Task:

Analyze given code where mixed indentation breaks execution. Use AI to fix it.

Bug: Mixed indentation

```
def func():
```

```
    x = 5
```

```
        y = 10
```

```
    return x+y
```

Expected Output : Consistent indentation applied.

Prompt:

Fix inconsistent indentation.

Code:

```
1  # Bug: Mixed indentation
2  def func():
3      x = 5
4      y = 10
5      return x+y
6  print(func())
```

Output(O/P)

```
15
```

Explanation:

Python requires consistent indentation.

Using spaces consistently fixes execution.

Question 8: Import Error – Wrong Module Usage

Task:

Analyze given code with incorrect import. Use AI to fix it.

Bug: Wrong import

import maths

print(maths.sqrt(16))

Expected Output: Corrected to import math

Prompt:

Fix incorrect module import.

Code:

```
1  # Bug: Wrong import
2  import math
3  print(math.sqrt(16))
```

Output(O/P)

```
4.0
```

Explanation:

Corrected Typo of Import of math module

Question 9: Unreachable Code – Return Inside Loop

Task:

Analyze given code where a return inside a loop prevents full iteration.

Use AI to fix it.

Bug: Early return inside loop

```
def total(numbers):
```

```
    for n in numbers:
```

```
        return n
```

```
print(total([1,2,3]))
```

Expected Output: Corrected code accumulates sum and returns after loop

Prompt:

Fix early return inside loop.

Code:

```
1  # Bug: Early return inside loop
2  def total(numbers):
3      sum_total = 0
4      for n in numbers:
5          sum_total += n
6      return sum_total
7  print(total([1,2,3]))
```

Output(O/P)

6

Explanation:

Return statement inside the loop stops after the first iteration.

Now it sums all elements before returning.

Question 10: Name Error – Undefined Variable

Task:

Task: Analyze given code where a variable is used before being defined.

Let AI detect and fix the error.

Bug: Using undefined variable

```
def calculate_area():  
    return length * width  
print(calculate_area())
```

Requirements:

- Run the code to observe the error.
- Ask AI to identify the missing variable definition.
- Fix the bug by defining length and width as parameters.
- Add 3 assert test cases for correctness.

Expected Output :

- Corrected code with parameters.
- AI explanation of the bug.

Successful execution of assertions.

Prompt:

Fix undefined variables by using parameters. Add 3 assert test cases to test correctness.

Code:

```
1  # Bug: Using undefined variable  
2  def calculate_area(length, width):  
3      return length * width  
4  
5  # Test cases  
6  assert calculate_area(5, 10) == 50  
7  assert calculate_area(3, 4) == 12  
8  assert calculate_area(7, 8) == 56  
9  
10 print(calculate_area(5, 10))
```

Output(O/P)

50

Explanation:

length and width were undefined.

Making them parameters fixes NameError.

Question 11: Type Error – Mixing Data Types Incorrectly

Task:

Analyze given code where integers and strings are added incorrectly. Let AI detect and fix the error.

Bug: Adding integer and string

```
def add_values():  
    return 5 + "10"
```

```
print(add_values())
```

Requirements:

- Run the code to observe the error.
- AI should explain why `int + str` is invalid.
- Fix the code by type conversion (e.g., `int("10")` or `str(5)`).
- Verify with 3 assert cases.

Expected Output #6:

- Corrected code with type handling.
- AI explanation of the fix. Successful test validation.

Prompt:

Fix int adding string error using conversion. Add 3 assert test cases to test correctness, explain in a docstring why `int + str` is invalid.

Code:

```
1  # Bug: Adding integer and string
2  def add_values():
3      """
4      Add two values together.
5
6      Note: In Python, the + operator cannot directly add an integer and string
7      because they are incompatible types. Strings require explicit conversion
8      to numeric types before arithmetic operations.
9      """
10     return 5 + int("10")
11
12     print(add_values())
13
14     # Test cases
15     assert add_values() == 15, "5 + 10 should equal 15"
16     assert isinstance(add_values(), int), "Result should be an integer"
17     assert add_values() > 10, "Result should be greater than 10"
```

Output(O/P)

15

Explanation:

Python cannot add int and str directly.
Type conversion solves the issue.

Question 12: Type Error – String + List Concatenation

Task:

Analyze code where a string is incorrectly added to a list.

Bug: Adding string and list

```
def combine():  
    return "Numbers: " + [1, 2, 3]
```

```
print(combine())
```

Requirements:

- Run the code to observe the error.
- Explain why str + list is invalid.
- Fix using conversion (str([1,2,3]) or " ".join()).
- Verify with 3 assert cases.

Expected Output:

- Corrected code
- Explanation
- Successful test validation

Prompt:

Fix string + list concatenation using conversion, Add 3 assert test cases to test correctness, explain in a docstring why str + list is invalid.

Code:

```
1  # Bug: Adding string and list
2  def combine():
3      """
4      Combine a string with a list representation.
5
6      Note: Direct concatenation of str + list is invalid in Python because
7      strings and lists are different types. We must convert the list to a
8      string representation using str() before concatenation.
9      """
10     return "Numbers: " + str([1, 2, 3])
11
12     print(combine())
13
14     # Test cases
15     assert combine() == "Numbers: [1, 2, 3]", "Basic concatenation failed"
16     assert isinstance(combine(), str), "Result should be a string"
17     assert "Numbers:" in combine() and "[1, 2, 3]" in combine(), "Both parts should be present"
```

Output(O/P)

```
Numbers: [1, 2, 3]
```

Explanation:

String cannot be directly added to the list.
Convert list to string first.

Question 13: Type Error – Multiplying String by Float

Task:

Detect and fix code where a string is multiplied by a float.

Bug: Multiplying string by float

```
def repeat_text():  
    return "Hello" * 2.5
```

```
print(repeat_text())
```

Requirements:

- Observe the error.
- Explain why float multiplication is invalid for strings.
- Fix by converting float to int.
- Add 3 assert test cases.

Prompt:

Fix string multiplied by float using conversion, Explain in a docstring why float multiplication is invalid for strings, Add 3 assert test cases.

Code:

```
1  # Bug: Multiplying string by float  
2  def repeat_text():  
3      """  
4          Repeat a string by converting the multiplier to an integer.  
5  
6          In Python, strings can only be multiplied by integers, not floats.  
7          Multiplying a string by a float raises a TypeError because the  
8          repetition count must be a whole number. We convert 2.5 to int(2)  
9          to fix this issue.  
10         """  
11     return "Hello" * int(2.5)  
12  
13 print(repeat_text())  
14  
15 # Test cases  
16 assert repeat_text() == "HelloHello", "Should repeat string 2 times (2.5 truncated to 2)"  
17 assert len(repeat_text()) == 10, "Result should have length 10"  
18 assert repeat_text()[0:5] == "Hello", "First 5 characters should be 'Hello'"
```

Output(O/P)

```
HelloHello
```

Explanation:

Strings can only be multiplied by integers.
Convert float to int before multiplication.

Question 14: Type Error – Adding None to Integer

Task:

Analyze code where None is added to an integer.

Bug: Adding None and integer

```
def compute():  
    value = None  
    return value + 10
```

```
print(compute())
```

Requirements:

- Run and identify the error.
- Explain why NoneType cannot be added.
- Fix by assigning a default value.
- Validate using asserts.

Prompt:

Fix None addition issue, Explain in a docstring why NoneType cannot be added, Validate using asserts.

Code:

```
1  # Bug: Adding None and integer  
2  def compute():  
3      """  
4      Computes a value.  
5  
6      NoneType cannot be added to integers because None represents the absence  
7      of a value and has no numeric representation. Python's type system prevents  
8      operations between incompatible types to avoid undefined behavior.  
9      """  
10     value = 0 # Fixed: Initialize with a valid integer instead of None  
11     return value + 10  
12  
13     result = compute()  
14     print(result)  
15  
16     # Validate using asserts  
17     assert result == 10, "Expected result to be 10"  
18     assert isinstance(result, int), "Expected result to be an integer"
```

Output(O/P)

10

Explanation:

NoneType cannot be added to int.

Assign default numeric value before operation.

Question 15: Type Error – Input Treated as String Instead of Number

Task:

Fix code where user input is not converted properly.

Bug: Input remains string

```
def sum_two_numbers():  
    a = input("Enter first number: ")  
    b = input("Enter second number: ")  
    return a + b  
print(sum_two_numbers())
```

Requirements:

- Explain why input is always string.
- Fix using int() conversion.
- Verify with assert test cases.

Prompt:

Fix input conversion issue using type conversion, Explain in docstring why input is always string, Verify with assert test cases.

Code:

```
1  # Bug: Input remains string  
2  def sum_two_numbers():  
3      """  
4          Sum two numbers from user input.  
5  
6          Why input() returns strings:  
7          The input() function always returns user data as a string type,  
8          regardless of what the user types. String concatenation occurs  
9          instead of numeric addition without explicit type conversion.  
10     """  
11     a = float(input("Enter first number: "))  
12     b = float(input("Enter second number: "))  
13     return a + b  
14  
15     result = sum_two_numbers()  
16     print(result)  
17  
18     # Test cases  
19     assert sum_two_numbers.__doc__ is not None, "Docstring missing"  
20     assert isinstance(result, float), "Result should be a float"  
21     assert result == 5.0, "Expected sum of 2.0 and 3.0 to be 5.0"
```

Output(O/P)

```
Enter first number: 2.0  
Enter second number: 3.0  
5.0
```

Explanation:

input() always returns string.
Convert to float or int before addition.